

Metabolic dysfunction-associated steatotic liver disease and its link to cancer.

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Metabolic-dysfunction associated steatotic liver disease (MASLD), formerly known as nonalcoholic fatty liver disease (NAFLD), is a growing global health concern with significant implications for oncogenesis. This review synthesizes current evidence on the association between MASLD and cancer risk, highlighting its role as a risk factor for both intrahepatic and extrahepatic malignancies. MASLD is increasingly recognized as a major cause of hepatocellular carcinoma (HCC), with its incidence rising in parallel with the prevalence of metabolic dysfunction. Furthermore, MASLD is associated with an elevated risk of various gastrointestinal cancers, including colorectal, esophageal, stomach, and pancreatic cancers. Beyond the digestive tract, evidence suggests that MASLD may also contribute to an increased risk of other cancers such as breast, prostate, thyroid, gynecological, renal and lung cancers. Understanding the mechanisms underlying these associations and the impact of MASLD on cancer risk is crucial for developing targeted screening and prevention strategies.

Roles of Activin A and Gpnmb in Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD).

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Metabolic dysfunction-associated steatotic liver disease (MASLD, formerly known as nonalcoholic fatty liver disease [NAFLD]) and metabolic dysfunction-associated steatohepatitis (MASH, formerly known as nonalcoholic steatohepatitis [NASH]) are leading chronic liver diseases, driving cirrhosis, hepatocellular carcinoma, and mortality. MASLD/MASH is associated with increased senescence proteins, including Activin A, and senolytics have been proposed as a therapeutic approach. To test the role of Activin A, we induced hepatic expression of Activin A in a murine MASLD/MASH model. Surprisingly, overexpression of hepatic Activin A dramatically mitigated MASLD, reducing liver steatosis and inflammation as well as systemic fat accumulation, while improving insulin sensitivity. Further studies identified a dramatic decrease in the lipid-associated macrophages marker glycoprotein NMB (Gpnmb) by Activin A, and Gpnmb knockdown in the same model produced similar benefits and transcriptional changes to Activin A expression. These studies reveal a surprising protective role for Activin A in MASLD and the potential for SASP proteins to have context-specific beneficial effects. Moreover, they implicate both Activin A and Gpnmb as potential therapeutic targets for this condition.

Advancements in metabolic-associated steatotic liver disease research: Diagnostics, small molecule developments, and future directions.

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Metabolic (dysfunction)-associated steatotic liver disease (MASLD), formerly known as nonalcoholic fatty liver disease, is a growing global health concern with no approved pharmacological treatments. At the same time, there are no standard methods to definitively screen for the presence of MASLD because of its progressive nature and symptomatic commonality with other disorders. Recent advances in molecular understanding of MASLD pathophysiology have intensified research on development of new drug molecules, repurposing of existing drugs approved for other indications, and an educated use of dietary

supplements for its treatment and prophylaxis. This review focused on depicting the latest advancements in MASLD research related to small molecule development for prophylaxis or treatment and diagnosis, with emphasis on mechanistic basis at the molecular level.

Current epidemiology of chronic liver disease.

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Chronic liver disease presents a significant global health burden, characterized by several etiologies, including metabolic dysfunction-associated steatotic liver disease (MASLD), alcohol-related liver disease (ALD), chronic hepatitis B virus infection, and chronic hepatitis C virus infection. This review explored current epidemiological trends and projections for each etiology, looking into their respective burdens and challenges. MASLD, formerly known as nonalcoholic fatty liver disease, is the most prevalent cause of chronic liver disease, and its global incidence and prevalence are steadily rising. ALD, fueled by increased alcohol consumption, is also on the rise, with concerning implications for future mortality rates. Chronic hepatitis B and C infections remain major public health concerns, particularly in specific regions of the world, necessitating concerted efforts for screening and treatment. The coronavirus disease 2019 (COVID-19) pandemic has impacted the epidemiology of chronic liver disease, exacerbating mortality rates and disrupting healthcare services. Mental health issues arising from the pandemic further complicate the treatment of chronic liver disease, making comprehensive healthcare strategies essential. Despite advancements in treatment, chronic liver disease continues to impose a substantial economic burden, emphasizing the importance of preventive measures and early intervention. In conclusion, ongoing surveillance and research efforts are crucial for understanding and addressing the evolving landscape of chronic liver disease. Comprehensive strategies that encompass prevention, screening, and treatment of its different etiologies are essential for mitigating its impact and improving patient outcomes.

[Research progress on the histological scoring system for nonalcoholic fatty liver disease].

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Non-alcoholic fatty liver disease (NAFLD) has replaced chronic hepatitis B as the most common chronic liver disease in China and has now been renamed metabolic dysfunction-associated fatty liver disease (MAFLD). The Brunt, the American NASH Clinical Research Network (NASH-CRN), the European Steatosis, Activity, and Fibrosis/Fatty Liver Inhibition of Progression (SAF/FLIP), and the Pediatric NAFLD are currently the four semi-quantitative grading systems for histological evaluation. This paper reviews these four scoring systems for the clinical selection of appropriate systems for diagnosis and prognosis assessment. This article is a review, and in order to coordinate the evaluation criteria of various scoring systems, the old name "NAFLD" is used.

Metabolic dysfunction-associated steatotic liver disease: Navigating terminological evolution, diagnostic frontiers and therapeutic horizon-an editorial exploration.

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Metabolic dysfunction-associated steatotic liver disease (MASLD), once known as non-alcoholic fatty liver disease (NAFLD), represents a spectrum of liver disorders characterized by lipid accumulation within hepatocytes. The redefinition of NAFLD in 2023 marked a significant reposition in terminology, emphasizing a broader understanding of liver steatosis and its associated risks. MASLD is now recognized as a major risk factor for liver cirrhosis, hepatocellular carcinoma, and systemic complications such as cardiovascular diseases or systemic inflammation. Diagnostic challenges arise, particularly in identifying MASLD in lean individuals, necessitating updated diagnostic protocols and investing in non-invasive diagnostic tools. Therapeutically, there is an urgent need for effective treatments targeting MASLD, with emerging pharmacological options focusing on, among others, carbohydrate and lipid metabolism. Additionally, understanding the roles of bile acid metabolism, the microbiome, and dietary interventions in MASLD pathogenesis and management holds promise for innovative therapeutic approaches. There is a strong need to emphasize the importance of collaborative efforts in understanding, diagnosing, and managing MASLD to improve physicians' approaches and patient outcomes.

[NAFLD renaming to MAFLD, MASLD: background, similarities, differences, and countermeasures].

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Nonalcoholic fatty liver disease (NAFLD) is a chronic metabolic stress liver injury due to chronic malnutrition in genetically susceptible individuals. With the prevalence of obesity, diabetes, and metabolic syndrome, NAFLD has become the world's number one chronic liver disease, and the flaws of forty years of exclusive disease diagnostic criteria and stigmatized disease terminology are becoming increasingly prominent. To this end, in the past three years, the international consensus group and the three major hepatology societies have suggested that NAFLD be renamed metabolism-associated fatty liver disease (MAFLD) and metabolism-associated steatosis liver disease (MASLD). Here is a description of the background of naming NAFLD, MAFLD, and MASLD, the similarities and differences of the three terms, the existing disputes, and our country's coping strategies.

FibroScan-aspartate transaminase: A superior non-invasive model for diagnosing high-risk metabolic dysfunction-associated steatohepatitis.

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Non-alcoholic fatty liver disease (NAFLD) with hepatic histological NAFLD activity score ≥ 4 and fibrosis stage $F \geq 2$ is regarded as "at risk" non-alcoholic steatohepatitis (NASH). Based on an international consensus, NAFLD and NASH were renamed as metabolic dysfunction-associated steatotic liver disease (MASLD) and metabolic dysfunction-associated steatohepatitis (MASH), respectively; hence, we introduced the term "high-risk MASH". Diagnostic values of seven non-invasive models, including FibroScan-aspartate transaminase (FAST), fibrosis-4 (FIB-4), aspartate transaminase to platelet ratio index (APRI), etc. for high-risk MASH have rarely been studied and compared in MASLD. To assess the clinical value of seven non-invasive models as alternatives to liver biopsy for diagnosing high-risk MASH. A retrospective analysis was conducted on 309 patients diagnosed with NAFLD via liver biopsy at Beijing Ditan Hospital, between January 2012 and December 2020. After screening for MASLD and the exclusion

criteria, 279 patients were included and categorized into high-risk and non-high-risk MASH groups. Utilizing threshold values of each model, sensitivity, specificity, positive predictive value (PPV), and negative predictive values (NPV), were calculated. Receiver operating characteristic curves were constructed to evaluate their diagnostic efficacy based on the area under the curve (AUROC). MASLD diagnostic criteria were met by 99.4% patients with NAFLD. The MASLD population was analyzed in two cohorts: Overall population (279 patients) and the subgroup (117 patients) who underwent liver transient elastography (FibroScan). In the overall population, FIB-4 showed better diagnostic efficacy and higher PPV, with sensitivity, specificity, PPV, NPV, and AUROC of 26.9%, 95.2%, 73.5%, 72.2%, and 0.75. APRI, Forns index, and aspartate transaminase to alanine transaminase ratio (ARR) showed moderate diagnostic efficacy, whereas S index and gamma-glutamyl transpeptidase to platelet ratio (GPR) were relatively weaker. In the subgroup, FAST had the highest diagnostic efficacy, its sensitivity, specificity, PPV, NPV, and AUROC were 44.2%, 92.3%, 82.1%, 67.4%, and 0.82. The FIB-4 AUROC was 0.76. S index and GPR exhibited almost no diagnostic value for high-risk MASH. FAST and FIB-4 could replace liver biopsy as more effectively diagnostic methods for high-risk MASH compared to APRI, Forns index, ARR, S index, and GPR; FAST is superior to FIB-4.

Vitamin E for people with non-alcoholic fatty liver disease.

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Non-alcoholic fatty liver disease (NAFLD), recently renamed metabolic dysfunction-associated steatotic liver disease (MASLD), is the most common liver disease worldwide, affecting an estimated 3 in 10 people. The available treatment is far from optimal. Diet and lifestyle changes to promote weight loss and weight loss maintenance are the basic management of NAFLD, but these are difficult to achieve and maintain. Vitamin E has shown beneficial effects on oxidative stress, which plays a major role in the pathogenesis of NAFLD. However, there is uncertainty about the effects of vitamin E for people with NAFLD. To evaluate the beneficial and harmful effects of vitamin E alone, or vitamin E in combination with other vitamins or minerals, versus placebo or no intervention in people with NAFLD. We used recommended Cochrane search methods. The latest search was performed on 2 February 2024. We included randomised clinical trials that compared vitamin E alone, or in combination with other vitamins or minerals, at any dose, duration, and route of administration, versus placebo or no intervention, in people with NAFLD of any age, sex, or ethnic origin. We included participants with imaging techniques or histology-proven NAFLD and minimal alcohol intake, and participants with steatohepatitis who had liver biopsies. Our critical outcomes were all-cause mortality, liver-related mortality, and serious adverse events. Our important outcomes were liver-related morbidity, health-related quality of life, non-serious adverse events, biochemical response, and imaging assessment of the degree of fatty liver. We used Cochrane's RoB 2 tool to assess risk of bias for each of the predefined outcomes. We used standard Cochrane methods. We used GRADE to assess the certainty of evidence. We included 16 randomised clinical trials involving 1066 paediatric and adult participants with NAFLD. Experimental groups received vitamin E alone (14 trials) or vitamin E in combination with vitamin C (2 trials). Control groups received placebo in 13 trials and no intervention in three trials. Daily dosages of oral vitamin E ranged from 298 international units (IU) to 1000 IU. Co-interventions were lifestyle and low-calorie diet interventions in 13 trials, ursodeoxycholic acid in one trial, unchanged diet and physical activity in one trial, and baseline treatments for type 2 diabetes in one trial. Nine trials had more than two intervention groups, but we used only the groups in which vitamin E alone or vitamin E in combination with vitamin C were compared with placebo or no intervention. In total, 7.9% (84/1066) of participants dropped out. Follow-up ranged from 2 months to 24 months. Vitamin E versus placebo or no intervention The effects of vitamin E versus

placebo or no intervention on all-cause mortality (risk ratio (RR) 3.45, 95% confidence interval (CI) 0.57 to 20.86; 3 trials, 351 participants; very low certainty evidence) and serious adverse events (RR 1.91, 95% CI 0.30 to 12.01; 2 trials, 283 participants; very low certainty evidence) are very uncertain. There were no data on liver-related mortality or liver-related morbidity. The effects of vitamin E versus placebo or no intervention on physical health-related quality of life (mean difference (MD) 0.74, 95% CI -0.52 to 2.01; 2 trials, 251 participants; higher scores indicate better quality of life; very low certainty evidence); psychosocial health-related quality of life (MD -0.57, 95% CI -4.11 to 2.97; 2 trials, 251 participants; higher scores indicate better quality of life; very low certainty evidence); and non-serious adverse events (RR 0.86, 95% CI 0.64 to 1.17; 2 trials, 283 participants; very low certainty evidence) are also very uncertain. There were no data on proportion of participants without a decrease in liver enzymes. Vitamin E likely slightly reduces serum alanine transaminase (ALT) (MD -9.29, 95% CI -13.69 to -4.89; 11 trials, 708 participants; moderate certainty evidence) and aspartate aminotransferase (AST) (MD -4.90, 95% CI -7.24 to -2.57; 11 trials, 695 participants; moderate certainty evidence) levels compared with placebo or no intervention. Vitamin E may slightly reduce serum alkaline phosphatase (ALP) levels (MD -5.21, 95% CI -9.88 to -0.54; 5 trials, 416 participants; very low certainty evidence), but the evidence is very uncertain. Vitamin E plus vitamin C versus placebo There were no data on all-cause mortality, liver-related mortality, serious adverse events, liver-related morbidity, health-related quality of life, and non-serious adverse events. The effects of vitamin E plus vitamin C on reducing serum ALT (MD -0.50, 95% CI -4.58 to 3.58; 2 trials, 133 participants; very low certainty evidence), AST (MD 0.09, 95% CI -3.39 to 3.57; 1 trial, 88 participants; very low certainty evidence), and gamma-glutamyl transferase (GGT) levels (MD 1.58, 95% CI -3.22 to 6.38; 1 trial, 88 participants; very low certainty evidence) are very uncertain. We identified three ongoing trials, and six trials are awaiting classification. Given the very low certainty evidence, we do not know if long-term treatment (18 months to 24 months) with vitamin E administered alone affects all-cause mortality, serious adverse events, quality of life, or non-serious adverse events in people with NAFLD when compared with placebo or no intervention. We found no data on liver-related mortality, liver-related morbidity, or proportion of participants without a decrease in liver enzymes. Vitamin E likely reduces ALT and AST slightly when compared with placebo, but whether this has any impact on the clinical course in people with NAFLD is unknown. The trials on vitamin E plus vitamin C did not report on all-cause mortality, liver-related mortality, serious adverse events, liver-related morbidity, health-related quality of life, or non-serious adverse events. Given the very low certainty evidence, we do not know the effects of vitamin E plus vitamin C on liver enzymes in people with NAFLD when compared with placebo. Three trials disclosed no external funding. Five trials were industry funded. Five trials were funded by organisations with no vested interests. Three trials did not provide any information on clinical trial support or sponsorship. Protocol: doi.org/10.1002/14651858.CD015033.

Metabolism-related signalling pathways involved in the pathogenesis and development of metabolic dysfunction-associated steatotic liver disease.

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Metabolic dysfunction-associated steatotic liver disease (MASLD) which formerly known as non-alcoholic fatty liver disease (NAFLD) is one of the causes of liver cirrhosis. Currently, a growing number of liver cirrhosis cases develop on the basis of MASLD, and the pathogenesis of MASLD remains unclear. This paper reviews the research progress on the involvement of different metabolism-related signalling pathways in the pathogenesis and development of MASLD.